

East West University
Department of Computer Science and Engineering
Course Code: CSE 207 Title: Data Structures
Course Instructor: Manowarul Islam
Date: 03.03.2022 Fall 2022
Class Test-1 Section: 5
Time: 1 hour 20 minutes Marks: 20

Note: Answer all questions. Figure in the right margin indicate marks.

1. Find the postfix equivalent of the following expression and evaluate the result. **6**
 $(4+8) * (7-5) / ((4-2) * (2+2))$

2. Suppose you have a circular queue of size 6 (1 to 6). Initially the value of the front and rear is 0. Now show the state of the queue if the following operation is performed on the queue. Also find the value of front and rear in each step. **4**

insert (10); insert (5), delete (), insert (9); insert (9); insert (7); insert (5); insert (8); insert (10); insert (11).

3. State the memory representation of single linked list with example. **3**

4. Write a simple procedure to count the number of elements in a linked list. **4**

5. Linked lists have several advantages over arrays. Mention some advantages. **3**